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Pathway-Priming of Intermediate Phases in FACs-based Wide-Bandgap Perovskites for Si Tandem Solar Cells

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Abstract

Wide-bandgap (WBG) perovskite solar cells (PSCs) have emerged as promising candidates for tandem applications, despite potential instability induced by halogen components. Formamidinium-cesium (FACs)-based perovskites represent as an intrinsically stable WBG system due to their low bromine-to-lead ratio and the absence of thermally unstable methylammonium. However, the performance of FACs-based WBG perovskites is highly dependent on achieving precise control over the formation dynamics. Here, we employ in-situ characterization to reveal that the key to this control lies in the evolution of the intermediate phase and propose a pathway-priming strategy. By strategically guiding the early formation of the $CsPbX_3$ intermediate phases, the strategy in turn templates and accelerates the subsequent transformation to the desired α-phase. The primed pathway yields a more uniform and complete phase transition, leading to enhanced film homogeneity. As a result, we demonstrate highly efficient 1.67 eV WBG PSCs with an efficiency exceeding 23%, along with excellent long-term operating stability maintaining 90% initial performance for 800 hours under continuous light stress. Furthermore, semitransparent WBG PSC achieve efficiencies of 20.91% (0.1 cm^2) and 19.67% (1.0 cm^2), corresponding to a four-terminal perovskite/silicon tandem solar cells with an efficiency of 31.04% (1.0 cm^2), highlighting their potential for high-performance tandem photovoltaics.

Keywords

Wide bandgap; Intermediate phase; Stability; Tandem solar cell.

Introduction

Perovskite solar cells (PSCs) with an ABX_3 (A = formamidinium (FA), methylammonium (MA), cesium (Cs); B = lead (Pb); X = iodide (I), bromide (Br)) structure have emerged as one of the most promising candidates for future photovoltaic technology, offering both remarkable power conversion efficiency (PCE) and significantly lower manufacturing costs[1,2]. In addition to single-junction applications, PSCs have enabled advances in tandem solar cells (TSCs), providing the potential to break through the Shockley-Queisser limit[3]. Perovskite/silicon (Si) TSCs are of great interest in both academic and industry recently, as Si-based solar cells dominate the photovoltaic industry and seek efficiency upgrades[4,5]. However, the large stability mismatch between Si and perovskite sub-cells poses a major commercialization barrier. In this regard, four-terminal (4T) TSCs present a more feasible solution, offering greater design flexibility of spectral matching and lifetime requirements[6].

For wide-bandgap (WBG) perovskites in tandem devices, compositions with high Br content ($AB(I_{1-x}Br_x)_3$, $x > 0.2$) exhibit susceptibility to light-induced halide segregation[7,8]. A-site Cs incorporation addresses both phase separation by allowing bandgap tuning with reduced Br, and thermal degradation by removing MA[9,10]. Nevertheless, how Cs influences the crucial early-stage crystallization process, specifically the formation and transformation of intermediate phases[11], remains a missing link in optimizing FACs-based WBG systems. Current FA-based systems encompass diverse intermediate phases, including solvated PbX_2, $AX{\cdot}PbX_2{\cdot}$solvent and δ_H-$FAPbX_3$ (A = FA, MA, X = I, Br)[12-14]. The incomplete transformation of the intermediate phase can easily lead to nanoscale phase impurities, which act as traps for carrier transportation as well as focal points for degradation[13,15-18]. Consequently, based on the nucleation-growth dynamics analysis, intentional modulation of the evolution process is crucial for efficient and stable FACs-based WBG PSCs.

This work introduces a pathway-priming strategy for the FACs-based WBG system. The metastable $CsPbX_3$ phases served as a crucial bridge between the initial precursor state and the final photoactive phase. We find that guanidinium iodide (GAI) is able to function as a priming agent by reducing the formation energy barrier of the $CsPbX_3$ intermediate, thus shifting the kinetic bottleneck to an earlier stage. This guided evolution promotes a swift and thorough transformation into the corner-sharing α-phase perovskite upon annealing. The optimized formation dynamics eliminates the phase impurities in the photoactive α-phase perovskite. Moreover, the homogeneity modulates internal stress/strain and enhances structural stability. As a result, the target 1.67 eV WBG PSCs achieved an improved PCE of 23.05% with exceptional operational stability, retaining 90% of initial performance after 800 hours under continuous 1-sun illumination. The target film shows excellent performance in inhibiting ion migration, remaining stable under 1.5-sun illumination in ambient atmosphere. Specifically, semitransparent WBG PSCs achieve efficiencies of 20.91% (0.1 cm^2) and 19.67% (1.0 cm^2). Furthermore, by stacking the semi-transparent PSCs (1.0 cm^2) with a silicon bottom solar cell, we achieved 4T perovskite/Si TSCs with a combined efficiency reaching 31.04%, showcasing the potential for tandem photovoltaic applications.

Results

To achieve a WBG system at low Br content, we specialize in perovskites composed of $FA_{0.8}Cs_{0.2}Pb(I_{0.8}Br_{0.2})_3$, in which the proportion of Br is limited to 20%. Based on the purpose of revealing the nucleation and crystallization process in FACs-based WBG perovskite systems, we try to illustrate it with a schematic diagram (**Figure 1a**) based on the in-situ photoluminescence (PL) measurements. For quantitative analysis of the evolution, we identified and analyzed the characteristic PL peaks**,** monitoring their spectral positions (**Figure 1b**) and intensities (**Figure 1c**) to elucidate the compositional and structural changes. Before the anti-solvent treatment, the precursor wet film was formed in a solvent-coordinated complex with no distinct PL signal. The analysis reveals that ethyl acetate (EA) antisolvent treatment promotes the formation of the characteristic ~698 nm emission peak. In order to identify the compositional of the intermediated phase, we established a reference in pure $CsPbI_3$ and $CsPb(I_{0.8}Br_{0.2})_3$ system (**Supplementary Fig. 1**), which indicated that the ~698 nm PL peak is belonged to the Cs-Pb-X system. Furthermore, X-ray diffraction (XRD) analysis confirms that the $CsPb(I_{0.8}Br_{0.2})_3$ film is a δ and α phase coexistence (**Supplementary Fig. 2a**). Therefore, the ~698 nm PL peak can be attribute to the photoactive $CsPbX_3$ phase. We perform XRD analysis for the $FA_{0.8}Cs_{0.2}Pb(I_{0.8}Br_{0.2})_3$ wet film, which showed a δ_O-$CsPbX_3$ XRD signal (**Supplementary Fig. 3**). At the same time, the wet film appears brown between yellow (δ phase) and black (α phase), which is also a visual mixture of the two phases (**Supplementary Fig. 2c**). Thus, the intermediate phase can be directed to the composition of $CsPbX_3$ (including both δ and α phase). The appearance and formation of the intermediate phase is marked as stage I. The subsequent annealing step triggering the redshift of the PL peak is marked as stage II. The stable state of the film after annealing is marked as stage III. To confirm the specific ion incorporation for the redshift of PL peak in the stage II, we compared the PL evolution of different halogen systems (**Supplementary Fig. 4**). The rapid redshift in the PL peak position during the thermal annealing in both Br-containing and pure I systems support the kinetic model of annealing-driven FA^+ final incorporation. Notably, this incorporation behavior of FA^+ is sensitive to the local chemical environment. The annealing process completes the final step of FA^+ from partial incorporation to complete homogenization in pure I system, while the intermediate phase strongly rejects FA^+ in Br-containing system.

From preliminary analysis, it can be seen that the metastable $CsPbX_3$ phase plays a key transitional role in the overall evolution of nucleation and growth. The $CsPbX_3$ serves as the intermediate phase during the phase-transition, which can be attributed to the rise in Cs content[19]. On this basis, we introduce the pathway-priming strategy, which relies on the targeted formation of the intermediate phase to preconfigure the system, thereby enabling an efficient and complete phase transition. Therefore, guanidinium iodide (GAI) is introduced as a priming agent. The addition of GA^+ gives rise to a much earlier initiation of the perovskite nucleation in stage I (**Figure 1d**). PL intensity evolution versus time further demonstrates the detailed kinetic enhancement (**Supplementary Fig. 5**). To be specific, the GA^+-incorporated system achieves a specific PL intensity in merely ~6 s, nearly twice as fast as the control group (~11 s). Based on the classical LaMer theory for monodispersed particle formation[20,21], it is suggested that the incorporation of GA^+ lowers the nucleation barrier of the intermediate phase so as to enter the stage of nucleation and fast growth earlier. To elucidate the kinetic role of GA^+, we deduce the reaction path of perovskite kinetics for both untreated and GA^+-treated systems (**Supplementary Fig. 6**). The observed acceleration in nucleation kinetics can be attributed to the reduction in the formation activation energy (E_a) with the GA^+ incorporation.

The accelerated stage I further promotes the phase transition process in stage II. The thermodynamic conditions provided by the annealing step allow FA^+ to be incorporated into the lattice rapidly, which corresponds to the phase transition of stage II. The slowdown stage II can be showed by the non-annealed PL tracking (**Supplementary Fig. 7a, b**). This transformation can be quickly completed without external forces, which makes it difficult for offline XRD to capture the specific intermediate phase of the wet film (**Supplementary Fig. 7c**). The priming pathway facilitates the subsequent incorporation of FA^+ into the crystal lattice and drives the complete transformation into photoactive α-phase. The detailed variation in the peak position (**Figure 2a**) and intensity (**Figure 2b**) of the PL during the annealing step indicates that the priming pathway accelerates the phase transition process. Moreover, the complete conversion of the metastable δ_O-$CsPbX_3$ phase has a decisive effect on the phase purity of the final film in stage III. In order to evaluate the consistency of the film, XRD measurements were performed, with **Figure 2c** and **Supplementary Fig. 8** revealing diffraction peaks corresponding to PbI_2 and δ_O-$CsPbX_3$[10]. Complementary grazing incidence wide-angle X-ray scattering (GIWAXS) measurements provided additional confirmation, with **Figure 2d** showing the line-cut profiles. The control films exhibited characteristic scattering features of δ-phase peaks at around 0.7 $Å^{-1}$[22] and PbI_2 peaks at 0.92 $Å^{-1}$[23]. These phase assignments were corroborated by 2D-GIWAXS patterns (**Figure 2e, f**), where diffraction rings at these q-values are marked by white arrows. The final state of phase purity serves as a critical indicator for unconverted interphase, directly indicating transformation completeness.

To reveal the possible mechanism of the pathway-priming strategy, Fourier transform infrared (FTIR) and nuclear magnetic resonance (NMR) spectroscopy were used to investigate the interactions among the perovskite precursors. FTIR analysis revealed a notable shift in the C=N stretching vibration of GAI from 1645 cm^{-1} to 1654 cm^{-1} upon interaction with PbI_2 (**Figure 2g**), indicating molecular interactions between these components. In contrast, negligible spectral changes were observed for the CsI-GAI system (**Supplementary Fig. 9**). The ^{1}H NMR spectra showed that the amino proton peak of GAI downshifted after mixing with PbI_2 (**Figure 2h** and **Supplementary Fig. 10**). The coordination bond between GA and I ions increases the electron cloud density of the imine group, resulting in the downshifts of ^{1}H-imine[24,25]. The ^{1}H NMR results also reveal that guanidinium (GA^{+}) competes with Pb^{2+} for coordination to DMSO molecules in the precursor solution (**Figure 2i**). The upfield shift of the DMSO signal from 2.54 ppm (pure DMSO) to 2.52 ppm with PbI_2 and to 2.51 ppm with GAI confirms that both Pb^{2+} and GA^{+} interact with the DMSO's S = O group. Crucially, the chemical shift moves to 2.53 ppm in the DMSO-GAI-PbI_2 system. This intermediate value indicates that neither interaction dominates, but both are simultaneously present and competing. The competitive coordination disrupts the $PbI_2{\cdot}(DMSO)_n$ complex, forming a less ordered and more labile $PbI_2{\cdot}(DMSO){\cdot}(GAI)$ hybrid complex. The precursor destabilization lowers the energy barrier for desolvation and initial nucleation. Consequently, the formation of $CsPbX_3$ intermediate nuclei is triggered earlier and more rapidly, as directly observed in the in-situ PL kinetics. This mechanism is further corroborated by in-situ PL of pure $CsPbI_3$ and $CsPb(I_{0.8}Br_{0.2})_3$ system with GAI, which exhibits a faster emergence of the characteristic emission peak during annealing (**Supplementary Fig. 1**).

Phase impurity directly affects the various properties of the film. The crystal structure of the perovskite film was studied by the grazing incident X-ray diffraction (GIXRD) technique. The measurement was performed at a 50 nm depth to further study the residual stress (**Figure 3a, b**). **Figure 3c** showed the corresponding fitted line of 2θ-$\sin^2\varphi$ for these perovskite films, which indicated the tensile strain on the surface of the film. The slope of the target film is estimated to be -0.089, which is smaller than that of the control film, indicating that the stress of the target film is smaller due to better homogeneity. The same conclusions can be mutually validated by other characterizations. Morphological symptom of the film surface was studied with the scanning electron microscope (SEM) and atomic force microscopy (AFM) (**Supplementary Fig. 11**). The target film exhibits a smaller root mean square (R_q) value of 2.62 nm than that of control film (3.78 nm), which is expected to provide a better interfacial contact between perovskite and functional layers. Additionally, the Kelvin probe microscopy (KPFM) was carried out to examine the surface potential of the perovskite film. As shown in the contact potential difference (CPD) mappings of the control film (**Figure 3d**), the film surface CPD had a relatively higher value[26]. The obvious color patches can be directly seen on the control film, which directly exhibits the unevenness of the surface composition. The CPD distribution (**Figure 3e**) of the target film was more concentrated and had a lower average value (**Supplementary Fig. 11c**), which can be accounted to the efficient passivation of positively charged defects, such as uncoordinated Pb^{2+} and iodine vacancy (V_I^{+})[27,28]. To further confirm the uniformity of the films, we conducted selected area PL mapping on both control and target perovskite films, examining the PL intensity on a microscale (**Figure 3f, g**). The control sample exhibits relatively low PL mapping intensity with individual non-uniform intensity areas, possibly due to the undesirable intermediate states. In contrast, the target films resulted in a substantial increase in intensity and uniform distribution.

To better understand the effect of phase purity on the carrier transportation and stability of the films, PL and TRPL were conducted. The steady-state PL peak of both films is located at 750 nm. The target perovskite film presents the stronger PL intensity compared to the control film (**Supplementary Fig. 12a**). The time-resolved photoluminescence (TRPL) measurements were performed to evaluate the dynamics of charge extraction. The enhanced lifetime of the target perovskite film indicates the lower trap states and suppressed non-radiative recombination in bulk perovskite films (**Supplementary Fig. 12b** and **Supplementary Table S1**). Continuous PL illumination was administered to evaluate the light-soaking stability of the films. As shown in **Figure 3h** and **i**, the control group showed a slight peak red-shift and a progressive increase in the full-width at half-maximum of the peak after 45 min of 1.5-sun intensity illumination, whereas the target film remained spectrally stable. The comparison of peak intensity and position before and after PL irradiation shows that the control sample has a light soaking effect. Therefore, GA^{+} not only enhances the film homogeneity but also improves the stability of the films. We speculate that this can be attributed to lattice stress release and hydrogen bonding of GA^{+} with adjacent halogen ions, which inhibits ion migration and contributes to the enhanced photostability[29,30]. In summary, the suppressed ion-migration and photoinduced phase segregation in WBG perovskite films are guaranteed to enhance overall device performance when applied in the full device stack.

The performance of the device is the verification of the effectiveness of the intermediate phase modulation strategy. We prepared single-junction WBG PSCs with an inverted architecture of ITO/NiO_x/4PADCB/WBG perovskite/$PC_{61}BM$/C_{60}/BCP/Ag (**Figure 4a** and **Supplementary Fig.**

13). **Figure 4b** shows the photocurrent density-voltage (*J–V*) curves of the target PSCs. The champion PCE of the target solar cell is 23.05% obtained by reverse voltage sweep with an open-circuit voltage (V_{OC}) of 1.26 V. The corresponding photovoltaic parameters analysis are summarized in **Supplementary Table S2**. Performance distributions of 20 cells for each type are shown in **Figure 4c** and **Supplementary Fig. 14** to verify the reproducibility. The target devices also demonstrated negligible hysteresis (**Figure 4b**) and stabilized power output (SPO) (**Supplementary Fig. 14a**), further highlighting the reliability.

Device-level observations can help to understand the contribution of GA^+ incorporation to the development of performance. The lower dark saturation current density of the target device indicates lower bulk defects in perovskite layer and suppressed non-radiative recombination[31] (**Supplementary Fig. 15a**). Moreover, **Supplementary Fig. 15b** shows the Mott–Schottky curves, from which it can be observed that the built-in voltage (V_{bi}) of the device increased from 1.14 to 1.16 V following the modification. GA^+ can promote charge carrier drift and extraction, thus leading to an increase in the fill factor (FF) of the device[32]. To quantitatively analyze the enhanced FF in target devices, we tested the V_{OC} as a function of illumination intensity to investigate the recombination mechanism during the device operation. As shown in **Supplementary Fig. 15c**, V_{OC} displayed a linear relationship with light intensity in a natural-logarithmic scale, and the target device had a smaller slope value of 1.32 k_BTq^{-1} compared with 1.41 k_BTq^{-1} of the control device, which suggests the suppressed trap-assisted recombination[33]. To further evaluate the non-radiative recombination, the EQE of electroluminescence (EQE_{EL}) was also measured (**Supplementary Fig. 15d**). The EL intensity of target devices is apparently stronger than that of pristine device, suggesting the retarded trap-assisted recombination[34].

The 1st derivative of the EQE spectrum and UV–vis absorption were measured to verify the bandgap (**Supplementary Fig. 16**), which is 1.67 eV[35]. The bandgap can achieve a good spectral match with the Si-based cell stack (**Figure 4d**), maximizing the use of input solar energy. Subsequently, on the basis of the optimized processes, semitransparent PSCs (0.1 cm^2) were fabricated and deliver an efficiency of 20.91% (**Supplementary Fig. 17**). EQE spectra of target devices are shown in **Figure 4e**. When the semitransparent PSC is extended to 1.0 cm^2 (19.67%) and further integrated with Si, a four-terminal tandem cell was obtained, delivering an efficiency of 31.04% (**Figure 4f**). The PCE for the 4T perovskite-Si TSCs is obtained by summing the operation PCE of subcells. The corresponding parameters of the optimized devices are included in **Supplementary Table S3**. To minimize the life mismatch of PSCs and silicon cells, the operational stability of the PSC was further evaluated by tracking the MPPT. As shown in **Figure 4g**, after MPPT for 800 h, the target device demonstrates good operational stability, retaining 90% of its initial PCE, while the control device reaches 80% of its initial PCE at ≈ 200 h. The comprehensive performance and stability assessments underscore the significant potential of 4T-TSCs for practical applications.

Discussion

In summary, we have decoded the formation dynamics of FACs-based WBG perovskites and established a pathway-priming strategy via $CsPbX_3$ intermediate phase control. Triggered by GA^+, the early formation of this intermediate ensures a complete phase transition, yielding superior films with minimal ion migration and efficient charge transport. Furthermore, we fabricate 4T perovskite/Si TSCs (1.0 cm^2) with a PCE of 31.04%, highlighting their potential for high-performance tandem photovoltaics. A deeper understanding of additive-driven intermediate phase modulation during perovskite formation will be pivotal for advancing efficient and stable perovskite/Si TSCs.

Methods

Materials

Lead iodide (PbI_2), Cesium Bromide (CsBr), Lead Bromide ($PbBr_2$), Cesium iodide (CsI), 1,3-Diaminopropane Dihydroiodide (PDAI) and CbzNaph (4PADCB) were purchased from TCI. Formamidinium iodide (FAI) was purchased from Greatcell Solar Materials. Phenyl-C61-butyric acid methyl-ester ($PC_{61}BM$) were purchased from Advanced Election Technology. C_{60}, guanidine hydroiodide (GAI) and bathocuproine (BCP) were purchased from Xi'an Polymer Light Technology Corp. Anhydrous dimethyl sulfoxide (DMSO), dimethylformamide (DMF) were purchased from Alfa Aesar. Isopropanol (IPA) was purchased from thermos scientific. Chlorobenzene (CB) was purchased form Sigma Aldrich. Ethyl alcohol was purchased from Aladdin. All chemicals were used without further purification.

Perovskite precursor

The perovskite with $FA_{0.8}Cs_{0.2}Pb(I_{0.8}Br_{0.2})_3$ precursor solutions were prepared by mixing the FAI, PbI_2, $PbBr_2$, CsBr and CsI in DMF/DMSO=3:1. The additive GAI is added to the precursor fluid at a molar ratio of 0.7%. Then the precursor solution was vibrated overnight by using a Vortex Shaker.

WBG perovskite device Fabrication

Patterned ITO substrates were cleaned using an ultrasonic bath in diluted detergent, deionized water, and IPA for 20 min in sequence. After drying, cleaned ITO substrates were treated for 30 min by UV-Ozone. The structure of the WBG perovskite cell is glass /ITO /NiO_X /4PADCB / $FA_{0.8}Cs_{0.2}Pb(I_{0.8}Br_{0.2})_3$ /PDAI /PCBM /C_{60} /BCP /Ag. NiO_X solution with 10 mg mL^{-1} in IPA: H_2O (3:1) was spin coated on ITO substrates at 3000 rpm for 25 s followed by annealing at 150 °C for 10 min. 4PADCB solution with 1 mg mL^{-1} in ethanol was spin coated on NiO_X films at 3000 rpm for 23 s followed by annealing at 100 °C for 10 min. The perovskite precursor solution was spin-coated on the top of substrates at 5000 rpm for 60 s, and 350 μL of EA was quickly dropped after 32 s, and then the film was annealed at 100 °C for 10 min. 40 μL of PDAI solution (1.5 mg mL^{-1} in IPA) was quickly dropped on the perovskite film and spin-coated at 4000 rpm. for 20 s, then annealed at 100 °C for 10 min. Subsequently, the $PC_{61}BM$ (10 mg mL^{-1} in CB) solution was spin-coated on top of the perovskite film at 3000 rpm for 30 s. In the following step, 20 nm of C_{60}, 5 nm of BCP and 90 nm of silver electrode were deposited via thermal evaporation at the evaporation rate of 0.1-0.5 Å/s.

Deposition of SnO_X layer

SnO_X films were deposited on substrates by an atomic layer deposition (ALD) technique in an ALD system (Wuhan Pudi). The reacting temperature during the deposition was kept at 100 °C, and TDMASn (at 60 °C) and H_2O (at 25 °C) were used as precursors for SnO_X fabrication. High-purity nitrogen (N_2, 99.999%) was used as carrier gas and purge gas. Chamber and process nitrogen flows were set to 12 and 12 sccm, respectively. A 12 nm thick SnO_X film was deposited by 100 cycles: TDMASn (pulse time 0.35 s, purge time 15 s) and H_2O (pulse time 0.12 s, purge time 20 s).

Semitransparent WBG perovskite device Fabrication

The fabrication of semitransparent-PSCs (ST-PSCs) was similar to that described for WBG-PSCs but with a few modifications. The structure of ST-PSCs is glass /ITO /NiO_X /4PADCB / $FA_{0.8}Cs_{0.2}Pb(I_{0.8}Br_{0.2})_3$ /PDAI /PCBM /C_{60} /ALD SnO_X /ITO /Ag grid. The C_{60} thickness was decreased to 15 nm. After that, ~12 nm of SnO_X was fabricated by atomic layer deposition technique. The top transparent electrode (ITO) was prepared at room temperature by radio frequency magnetron sputtering. ITO electrodes were sputtered at 80 W with Ar/O_2 flow rates of 20:0.2 sccm for 1800s. The base pressure was 8×10^{-4} Pa and the working pressure was maintained at 0.5 Pa. Then, a 70 nm-thick Ag grid was thermally evaporated at a rate of 1 Å/s through a shadow mask.

Characterization

Perovskite: The GIWAXS measurements were performed at BL02U2 beamline of the Shanghai Synchrotron Radiation Facility (SSRF). Grazing incidence X-ray diffraction (GIXRD) and X-ray diffraction (XRD) patterns were acquired in air by using a Rigaku Smartlab with Cu Kα (1.5418 Å) radiation in the 2θ range of 2-60° at a scanning rate of 15° min^{-1}. The photoluminescence (PL) spectra and time-resolved PL (TRPL) were performed using an Edinburgh Instrument FLS1000 system applying a 400 nm laser as the excitation source. The scanning electronic microscope (SEM) images were obtained from Apreo2 S Lovac field emission SEM at an acceleration voltage of 3 kV and a current of 50 pA. The atomic force microscopy (Anasys Instruments) was employed to probe the spatial distribution of perovskite films. The surface potential of the films was collected from the kelvin probe atomic force microscope (KPFM). The absorbance of the films was attained from UV-vis spectroscopy (HITACHI, UH5700), employing an optical range from 300 to 850 nm in air.

Devices: The current density versus voltage (*J-V*) characteristics of the PSCs were tested using a Keithley 2420 source meter under AM 1.5G one-sun illumination (100 mW·cm^{-2}) which was produced by a solar simulator (Sol3A Class AAA, Oriel, Newport, USA) at room temperature. A standard reference silicon cell (91150-KG3, Newport, USA) was used to calibrate the light intensity. The metal masks with an area of 0.1 cm^2 and 1.0 cm^2 was employed to determine the active area of the PSCs. The reverse scan range is from 1.3 V to -0.1 V and the forward scan range is from 0 V to 1.3 V. The current voltage scan speed is 0.2 V/s and dwell time is 20 ms. The testing environmental condition is in the ambient air with relative humidity of 60-80%. The incident photon to converted electron efficiency (IPCE) measurement was conducted to obtain the external quantum efficiency (EQE) spectra with a range from 300 to 850 nm using EQE system (IQE 200B, Newport). The stability of perovskite devices is evaluated under air conditions with 60-80% RH. The testing involves encapsulated samples (using glass-glass encapsulation with epoxy resins) to isolate the active layer from the environment. The electroluminescence (EL) spectra were characterized by a Keithley 2420 source meter and integrating sphere connected to a spectrophotometer (QE65Pro).

For PL and TRPL measurements, the decay lifetime was obtained by fitting the formula of $f(t) = A_1 exp^{-t/\tau_1} + A_2 exp^{-t/\tau_2} + B$ using biexponential decay model. B, A_1, and A_2 are constants associated with baseline offset and the contributions of fast (τ_1) and slow (τ_2) segments, respectively. The average carrier lifetime (τ_{ave}) can be determined from the equation of $\tau_{ave} = \frac{\sum A_i \tau_i^2}{\sum A_i \tau_i}$.

For Mott-Schottky measurements, the relationship between the capacitance and voltage is described by the M-S equation of $C^{-2} = \frac{2(V_{bi}-V)}{A^2 q \varepsilon \varepsilon_0 N_A}$, where V, A, q, ε, ε_0, N_A parameters are applied voltage, active area of device, charge of electron, relative dielectric constant, vacuum permittivity and doping density of perovskite, respectively. Based on this formula, the built-in potential could be evaluated.

For the measurement of open-circuit voltage under different light intensities, the relationship between open-circuit voltage and light intensity could be described by the formula of $qV_{OC} = E_g + n_{ID}k_BTln\frac{I}{I_0}$, where q, k_B, T, I, E_g stand for electron charge, the Boltzmann constant, the absolute temperature, light intensity and optical bandgap of perovskite, respectively.

Data availability

All data supporting the results of this study are provided in this article and its supplementary information. Any other information can be requested from the corresponding author. Source data are provided with this paper.

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Author Contributions Statement

L.Q., W.L. and P.H. supervised this work. N.H. conceived the ideas and designed the experiments. J.F. provided the GIWAXS measurements. X.W. participated in the fabrication of the semi-transparent PSCs. S.H. and S.L. conducted the GIXRD measurements. G.X. and L.G. provided valuable suggestions for the FTIR and NMR tests. H.P. and X.L. provided valuable suggestions for the experiments. X.Z., F.L., Z.Y. and P.Z. participated in the fabrication of the 4T perovskite/Si TSCs. N.H. wrote the manuscript and all authors reviewed the paper.

Competing Interests Statement

The authors declare no competing financial or non-financial interests.

Figure Legends

Figure 1. Pathway-priming of intermediate phases of perovskite. (a) Schematic illustration of the evolution of perovskite. The track of PL (b) peak position and (c) intensity over time for perovskite films during spinning coating and annealing stage. The blank interval region represents

the lack of a distinct characteristic peak. (d) In-situ PL measurements of control film and target film.

Figure 2. Revelation of the phase impurity of the perovskite film. The evolution of PL (a) peak position and (b) intensity over time for the control and target perovskite films during the annealing step. (c) XRD diffractions of perovskite films. (d) GIWAXS line-cut profiles (out-of-plane) of perovskite films. GIWAXS images of (e) control and (f) target perovskite film. (g) FTIR spectra to reveal the interactions. (h) ^{1}H NMR spectra of GAI, GAI & CsI, GAI & PbI_2. (i) ^{1}H NMR spectra of DMSO & GAI, DMSO & PbI_2, DMSO & GAI & PbI_2 and DMSO; the solvent is dimethyl sulfoxide-d6.

Figure 3. Property analysis of the film. GIXRD spectra of (a) control film and (b) target film with different instrumental tilt angles of φ values in the depth of 50 nm. (c) Linear fit of 2θ-$\sin^2\varphi$. KPFM images of (d) control film and (e) target film. Scale bars represent 500 nm. PL mapping images of (f) control film and (g) target film. Scale bars represent 20 μm. In-situ time-dependent PL evolution for (h) control film and (i) target film under ≈ 1.5-sun illumination for 45 min in ambient atmosphere while ageing. The sample stack is glass/ITO/HTL/perovskite.

Figure 4. Photovoltaic performance and device stability. (a) The device configuration of the single-junction WBG PSC (not to scale), where the SAM refers to self-assembled molecular layer. (b) *J*–*V* curves of the champion PSCs with antireflection film. (c) Statistics of the photovoltaic metrics (PCE, V_{OC}) of the control and target single-junction WBG PSCs. (d) A schematic illustration of the four-terminal perovskite/Si tandem solar cell (not to scale). (e) External quantum efficiency (EQE) spectra. (f) *J*–*V* characteristics of the semitransparent perovskite device (1.0 cm^2), stand-alone Si and filtered Si cells. (g) The maximum power point tracking (MPPT) stability of the encapsulated control and target devices under simulated one-sun illumination.

Editorial summary:

Wide-bandgap perovskites offer promise for tandems but suffer from formation-pathway instabilities. Huang et al. use in-situ analysis to guide $CsPbX_3$ intermediate evolution, enabling uniform α-phase formation and high-efficiency, stable devices.

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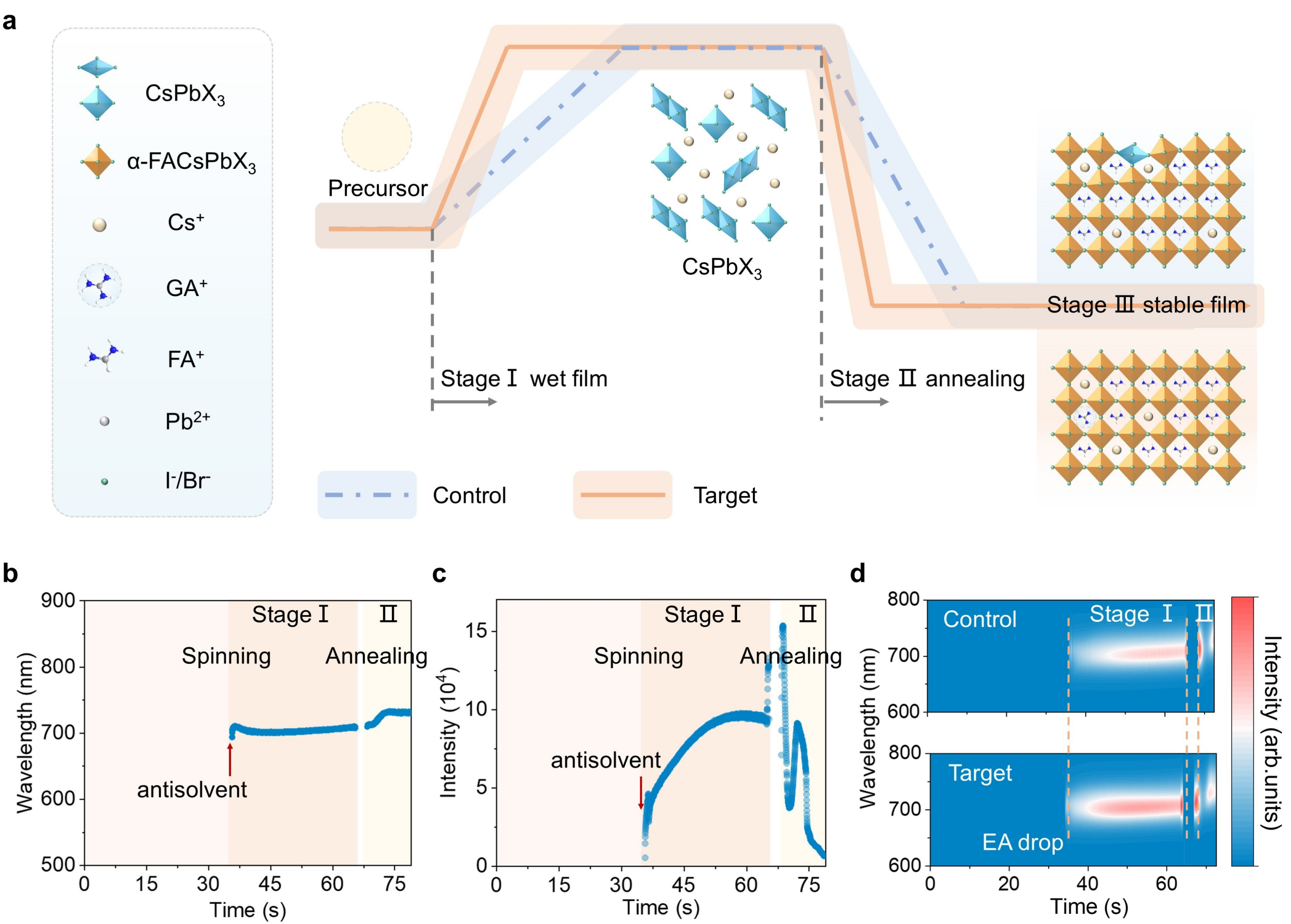
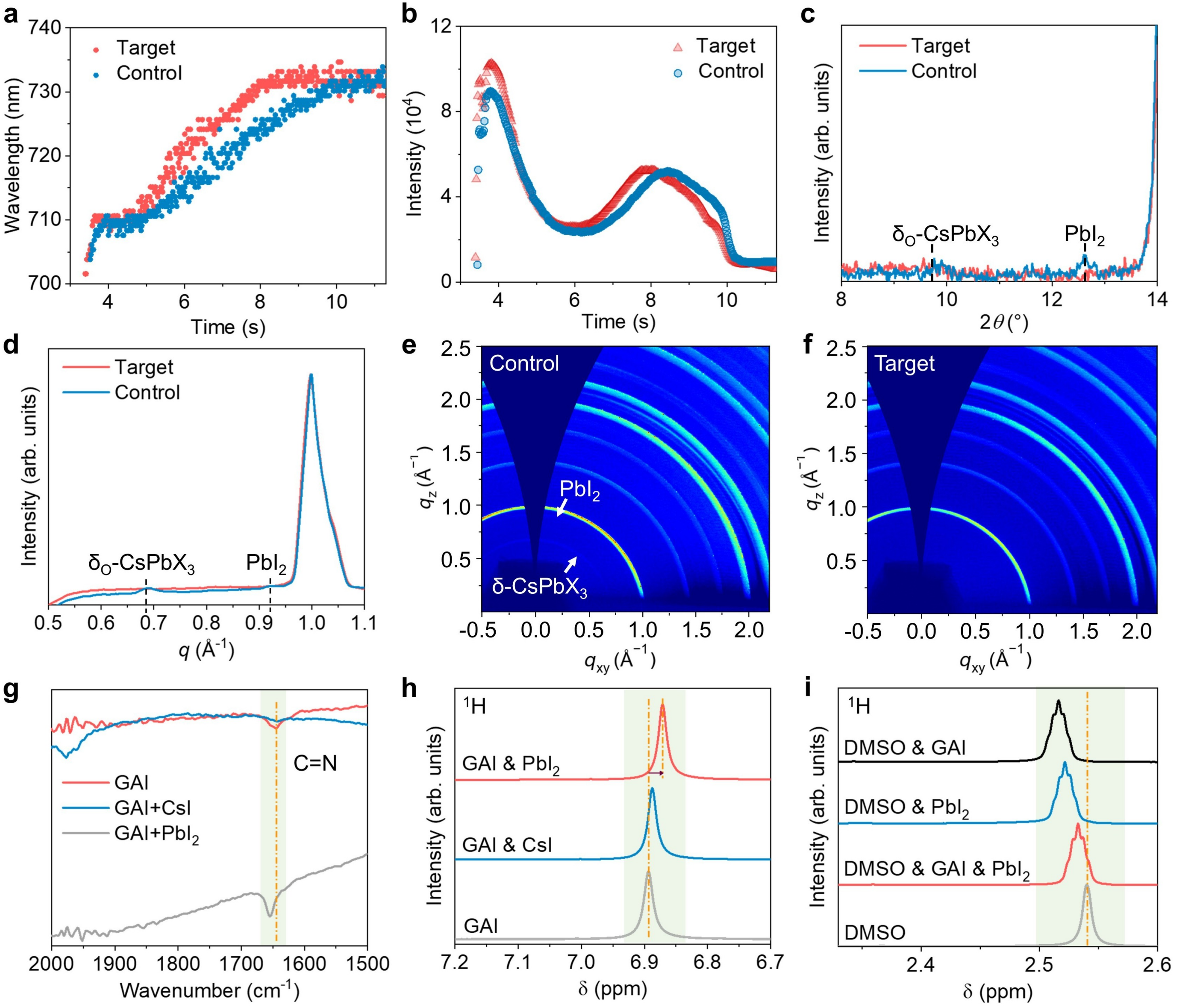
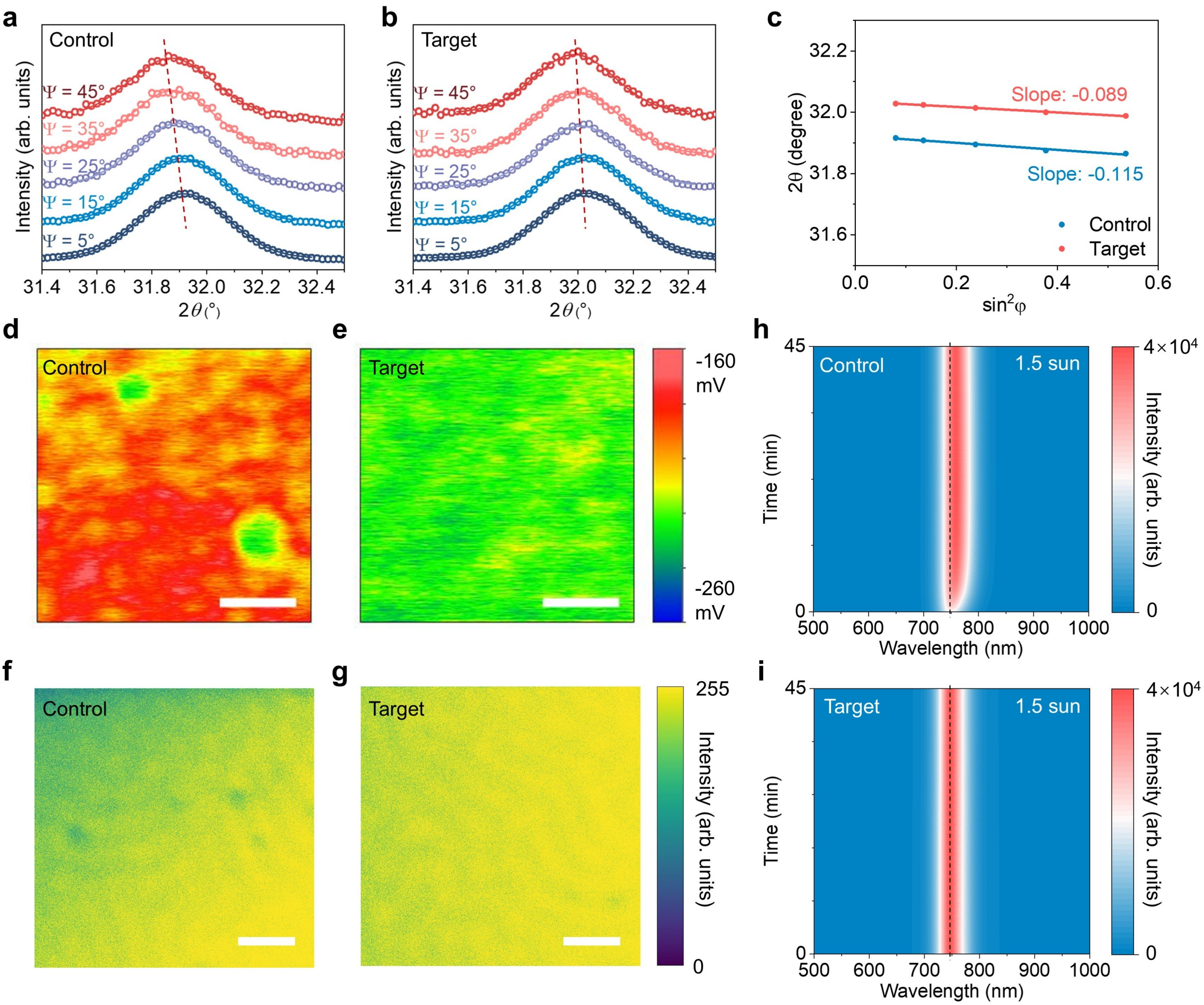
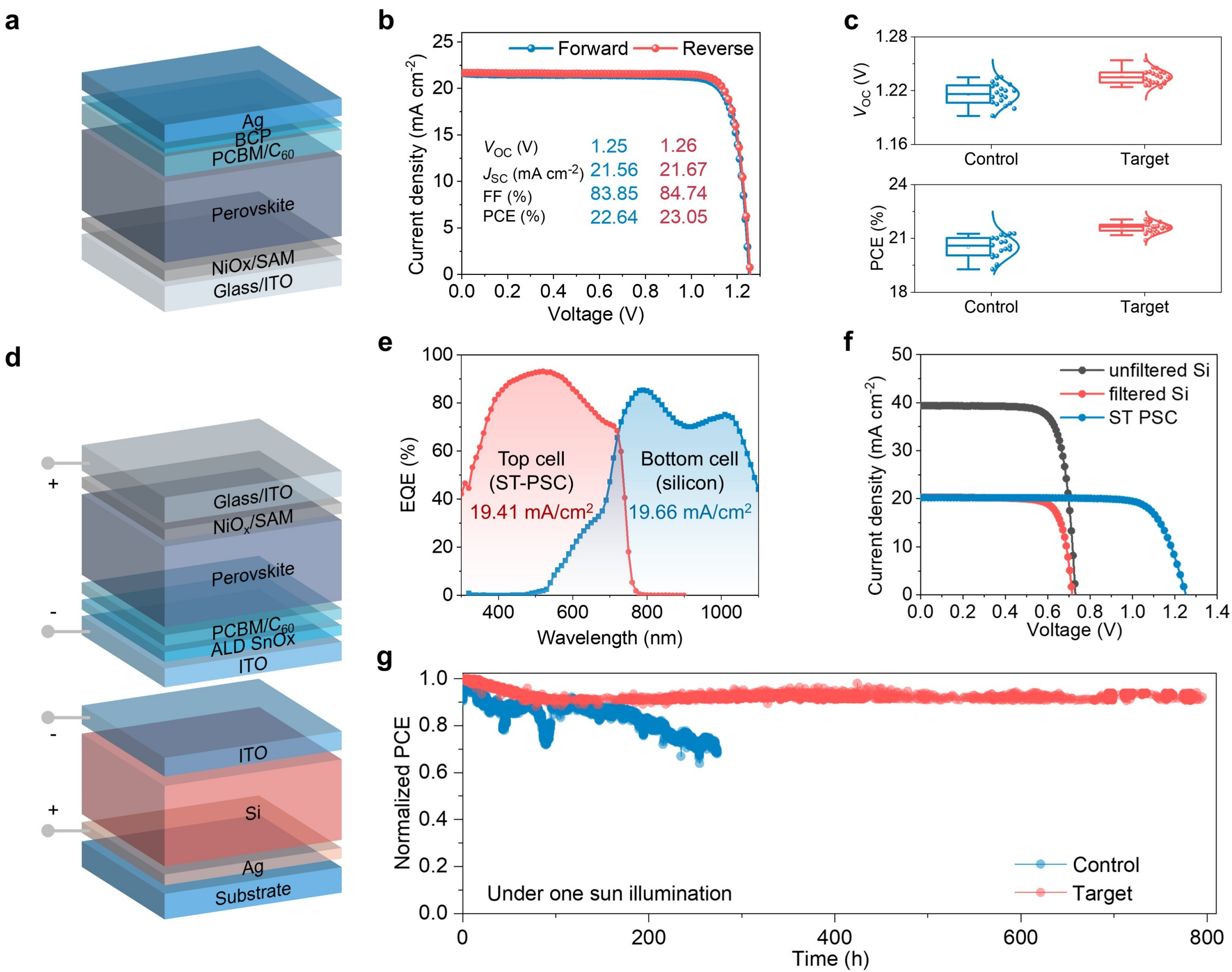

a
$CsPbX_3$
α-$FACsPbX_3$
Cs^+
GA^+
FA^+
Pb^{2+}
I^-/Br^-
Precursor
$CsPbX_3$
Stage Ⅰ wet film
Stage Ⅱ annealing
Stage Ⅲ stable film
Control
Target
b
Stage Ⅰ
Ⅱ
Spinning
Annealing
antisolvent
Wavelength (nm)
Time (s)
c
Stage Ⅰ
Ⅱ
Spinning
Annealing
antisolvent
Intensity (10^4)
Time (s)
d
Control
Target
Stage Ⅰ
Ⅱ
EA drop
Wavelength (nm)
Time (s)
Intensity (arb.units)


a
Target
Control
Wavelength (nm)
Time (s)
b
Target
Control
Intensity (10^4)
Time (s)
c
Target
Control
Intensity (arb. units)
δO-CsPbX3
PbI2
2θ (°)
d
Target
Control
Intensity (arb. units)
δO-CsPbX3
PbI2
q (Å-1)
e
Control
PbI2
δ-CsPbX3
qz (Å-1)
qxy (Å-1)
f
Target
qz (Å-1)
qxy (Å-1)
g
GAI
GAI+CsI
GAI+PbI2
C=N
Wavenumber (cm-1)
h
1H
GAI & PbI2
GAI & CsI
GAI
Intensity (arb. units)
δ (ppm)
i
1H
DMSO & GAI
DMSO & PbI2
DMSO & GAI & PbI2
DMSO
Intensity (arb. units)
δ (ppm)



a
Control
Ψ = 45°
Ψ = 35°
Ψ = 25°
Ψ = 15°
Ψ = 5°
Intensity (arb. units)
31.4 31.6 31.8 32.0 32.2 32.4
2θ(°)
b
Target
Ψ = 45°
Ψ = 35°
Ψ = 25°
Ψ = 15°
Ψ = 5°
Intensity (arb. units)
31.4 31.6 31.8 32.0 32.2 32.4
2θ(°)
c
32.2
32.0
31.8
31.6
2θ (degree)
Slope: -0.089
Slope: -0.115
Control
Target
0.0 0.2 0.4 0.6
sin²φ
d
Control
e
Target
-160 mV
-260 mV
h
45
0
Control
1.5 sun
Time (min)
500 600 700 800 900 1000
Wavelength (nm)
4×10⁴
0
Intensity (arb. units)
f
Control
g
Target
255
0
Intensity (arb. units)
i
45
0
Target
1.5 sun
Time (min)
500 600 700 800 900 1000
Wavelength (nm)
4×10⁴
0
Intensity (arb. units)


a
Ag
BCP
PCBM/C60
Perovskite
NiOx/SAM
Glass/ITO
b
Forward
Reverse
Current density (mA cm-2)
Voltage (V)
VOC (V) 1.25 1.26
JSC (mA cm-2) 21.56 21.67
FF (%) 83.85 84.74
PCE (%) 22.64 23.05
c
VOC (V)
PCE (%)
Control
Target
d
Glass/ITO
NiOx/SAM
Perovskite
PCBM/C60
ALD SnOx
ITO
Si
Ag
Substrate
e
EQE (%)
Top cell (ST-PSC)
19.41 mA/cm2
Bottom cell (silicon)
19.66 mA/cm2
Wavelength (nm)
f
unfiltered Si
filtered Si
ST PSC
Current density (mA cm-2)
Voltage (V)
g
Normalized PCE
Control
Target
Under one sun illumination
Time (h)